TASK#01

Iris Species Classification Using Machine Learning

Submitted By:

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1. Submission Links (Replace Placeholders)

Google Colab Notebook Link:
https://colab.research.google.com/drive/1P7ouP5H-_pLDmpJur6r2fdp_UNH_wn3s?usp=sharing

GitHub Repository Link:
https://github.com/ahsia931-source/Iris-classification-species-prediction

2. Project Overview

This project builds a multiclass classification model to predict Iris flower species using four classic botanical features: sepal length, sepal width, petal length, and petal width. Exploratory Data Analysis (EDA) was performed to understand distributions, outliers, and class separability. Three machine learning algorithms were trained and compared: K-Nearest Neighbors (KNN), Logistic Regression, and Decision Tree. The dataset was split into 80% training and 20% testing data using stratified sampling.

3. 1-Page Executive Summary

The objective of this project is to classify iris flowers into one of three species—Iris-setosa, Iris-versicolor, and Iris-virginica—based on four numerical measurements: sepal length, sepal width, petal length, and petal width. The dataset contains 150 observations and is a well-known benchmark for classification tasks.
Exploratory Data Analysis (EDA) was carried out using descriptive statistics, boxplots, histograms, a pairplot, a class count plot, and a scatter plot. These visualizations helped assess the distribution of each feature and the degree of separation between the three species. In particular, petal length and petal width showed strong discriminatory power.

Three machine learning models were trained and evaluated: K-Nearest Neighbors (KNN), Logistic Regression, and Decision Tree. The target variable was label-encoded and the data was split into training and testing sets with an 80:20 ratio using stratification to preserve class balance. Performance was measured using accuracy, precision, recall, F1-score, and confusion matrix.
Based on the notebook results, KNN achieved the highest test accuracy of 100%, Logistic Regression achieved 96.67%, and Decision Tree achieved 93.33%. Although the notebook saves Logistic Regression as the final model file (model.pkl), the best-performing model according to test accuracy is KNN. For full consistency, you may optionally change the saved model to KNN before final submission.
Overall, the project successfully demonstrates an end-to-end machine learning workflow: data loading, EDA, visualization, model training, model comparison, evaluation, model persistence using Joblib, and example inference.

4. Key Results Extracted from Notebook

Dataset shape: 150 rows × 5 columns

Train/Test Split: X_train = (120, 4), X_test = (30, 4), y_train = (120,), y_test = (30,)

KNN Accuracy: 1.00

Logistic Regression Accuracy: 0.9667

Decision Tree Accuracy: 0.9333

Saved model in notebook: model.pkl (currently saves Logistic Regression)

Inference example output in notebook: Predicted species: [0] (0 corresponds to Iris-setosa)

5. Important Recommendation Before Submission

Your notebook currently saves Logistic Regression as best_model, but KNN achieved the highest accuracy in your output. If you want the submission to be perfectly aligned with the requirement 'Save best model', update this line:
best_model = log_reg
to:
best_model = knn
Then re-run the save cell so model.pkl truly contains the best-performing model.

6. Screenshots from Notebook Outputs

The following figures were extracted from your uploaded notebook outputs and can be used as report screenshots.

Figure 1: Pair Plot showing class separability

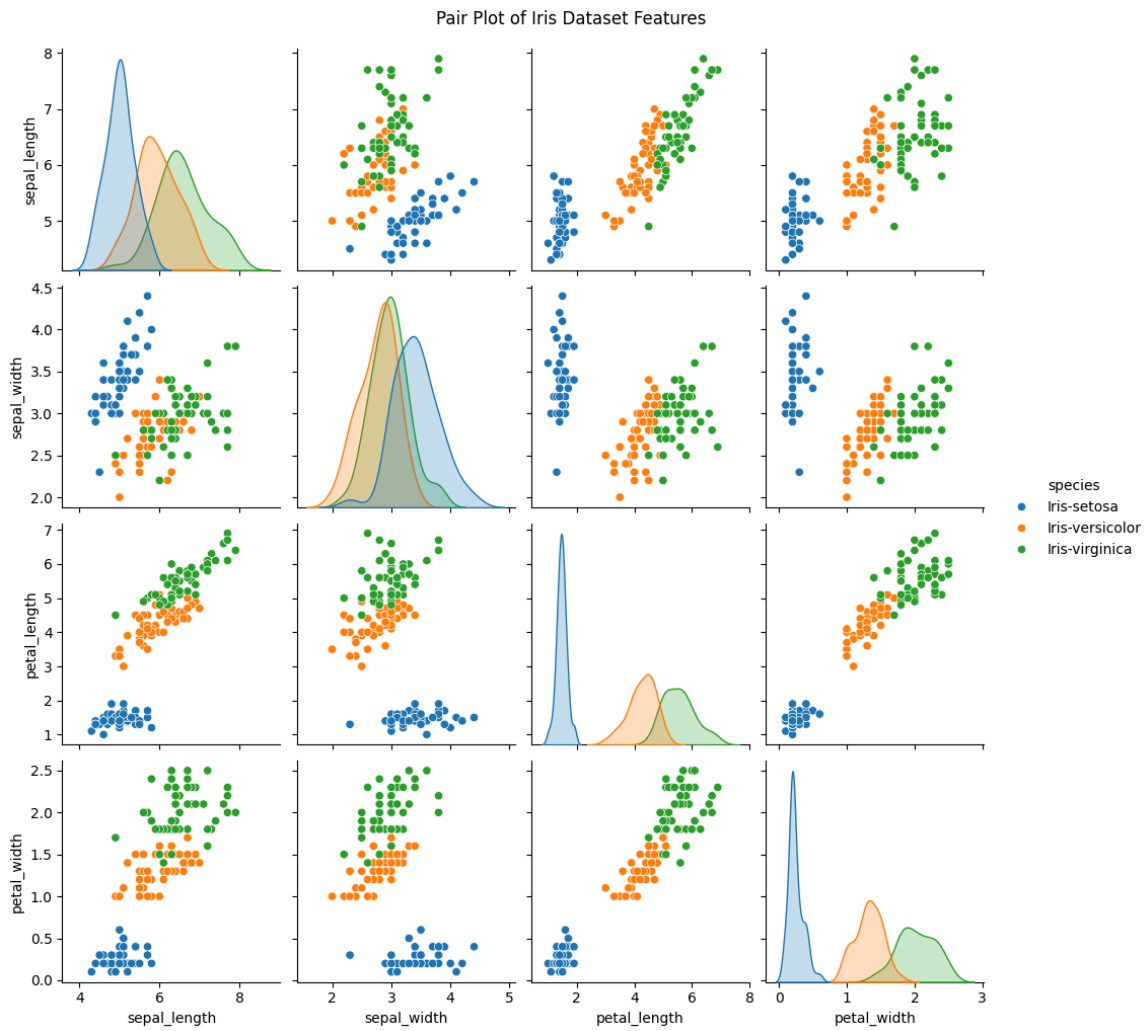


Figure 2: Count Plot of Iris species

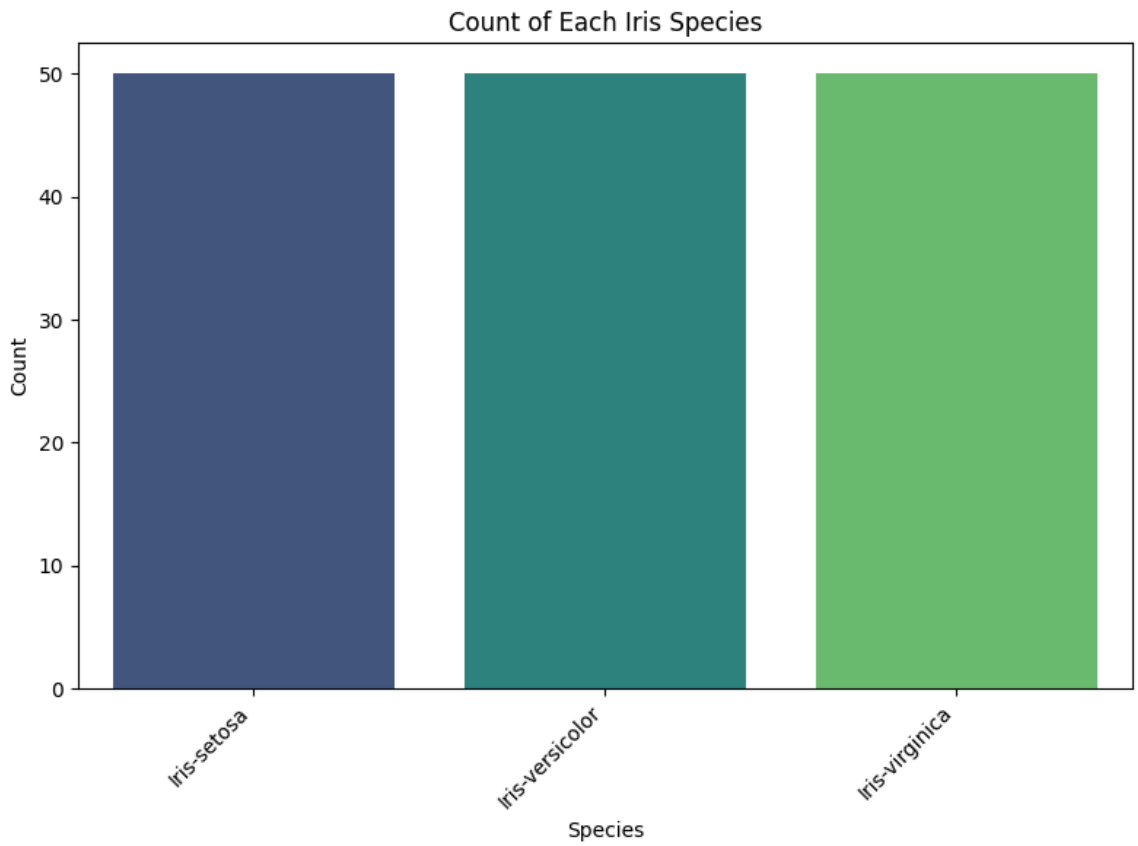


Figure 3: Scatter Plot (Sepal Length vs Sepal Width)

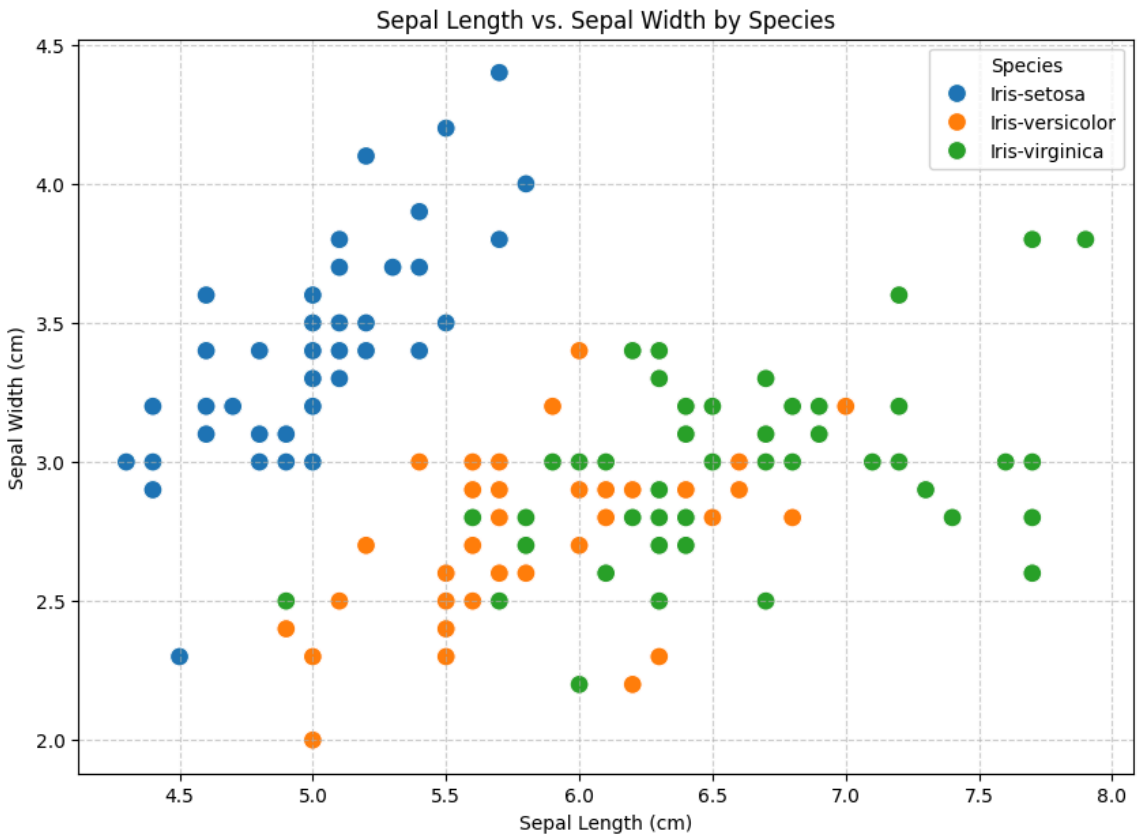


Figure 4: Example Boxplot from EDA

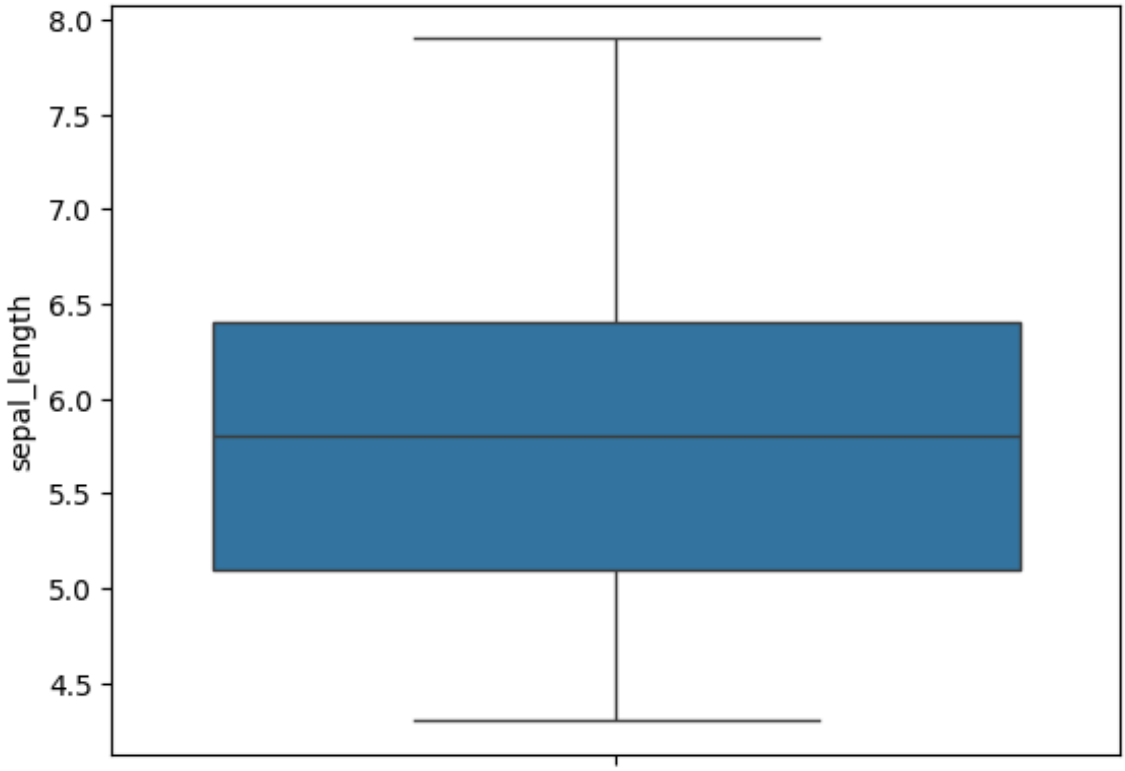
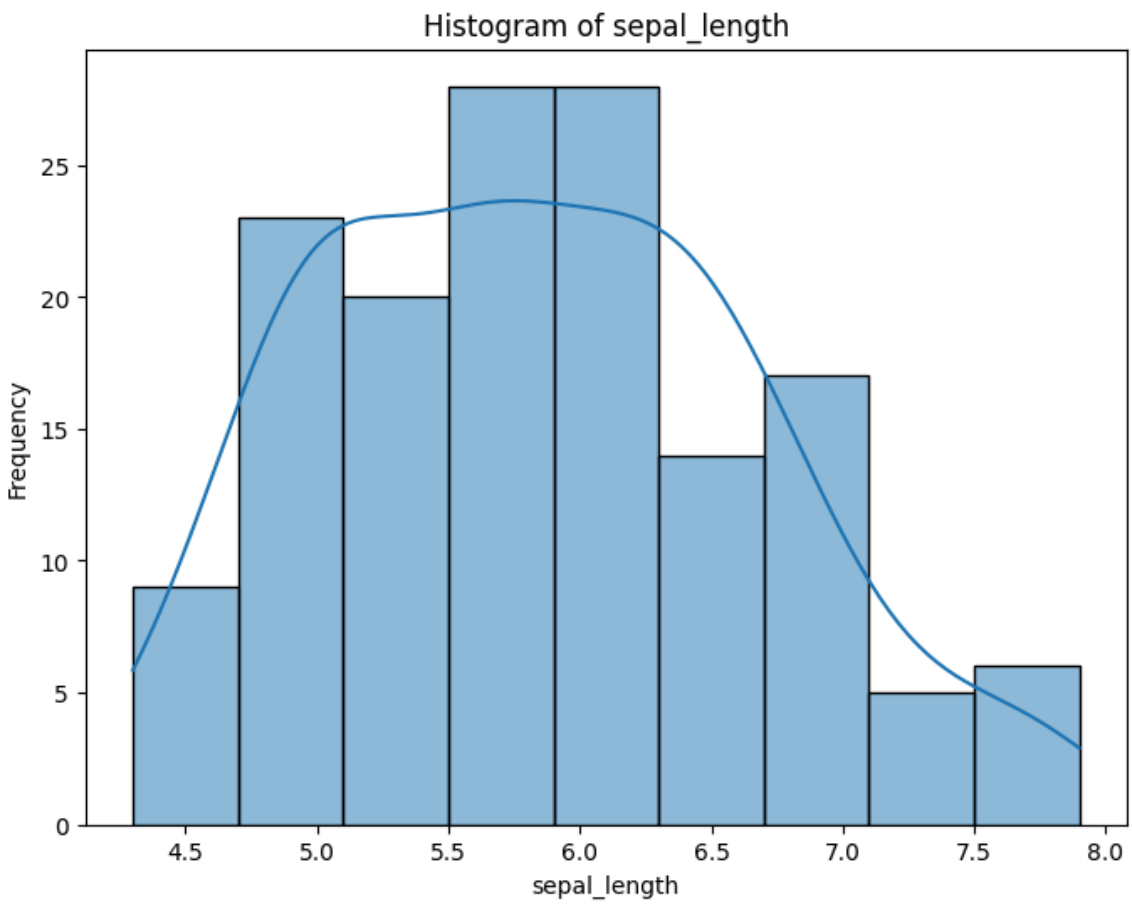


Figure 5: Example Histogram from EDA


Histogram of sepal_length
Frequency
sepal_length
0
5
10
15
20
25
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0